

Table 1: Correlation (Kendall's Tau) Between NLGCN Model and SIR Spreading Capacity (Light Green Highlight = Test Dataset)			
Network Dataset	Top 10% Correlation	Top 20% Correlation	Top 100% (All)
Budapest.txt	0.8101	0.7816	0.7815
C_elegans.txt	0.9360	0.8387	0.9152
E.coli.edge	0.8164	0.8134	0.8743
Human12a.edge	0.5668	0.6136	0.8086
NewSpain_18c_travelmap.txt	0.2987	0.4292	0.3755
dolphins.txt	0.8667	0.8485	0.8394
facebook_combined.txt	0.6196	0.7022	0.7060
karate.txt	0.3333	0.8667	0.8765
mammalia-voles-bhp-trapping.edges	0.6633	0.5629	0.7476
synthetic_test_realworld.txt	0.7968	0.7546	0.8476
US_airports.txt	0.9130	0.9288	0.8677
cargoshipsBB.txt	0.6688	0.7738	0.9042
carrib.txt	0.9638	0.9422	0.8932
cypedge.txt	0.7333	0.6667	0.9282
netscience.mtx	0.6889	0.4180	0.7909
open_flights.txt	0.9429	0.9287	0.8230
synthetic_test_sf_1000.txt	0.8040	0.7681	0.8365
synthetic_test_sf_2000.txt	0.7580	0.7333	0.8257
synthetic_test_sf_250.txt	0.8533	0.8612	0.8463
synthetic_test_sf_500.txt	0.8090	0.8053	0.8324
synthetic_test_sf_5000.txt	0.7337	0.7247	0.8111
synthetic_sf_100.txt	0.8667	0.8737	0.8403
synthetic_sf_1000.txt	0.8727	0.8375	0.8390
synthetic_sf_1500.txt	0.8375	0.7915	0.8378
synthetic_sf_2000.txt	0.8115	0.8014	0.8397
synthetic_sf_250.txt	0.8600	0.8465	0.8677
synthetic_sf_2500.txt	0.8061	0.7860	0.8301
synthetic_sf_3000.txt	0.8097	0.7529	0.8288
synthetic_sf_4000.txt	0.7830	0.7228	0.8199
synthetic_sf_500.txt	0.8906	0.8283	0.8416
synthetic_sf_850.txt	0.8784	0.8321	0.8466
synthetic_fragmented_1.txt	0.6667	0.5714	0.8978
synthetic_fragmented_2.txt	0.6000	0.6889	0.9059
synthetic_realworld_1.txt	0.6424	0.7531	0.8343
synthetic_realworld_2.txt	-0.0182	0.1909	0.2784

Table 2: Correlation (Kendall's Tau) Between Traditional Unweighted Centrality Measures and NLGCN Model (Light Green Highlight = Test Dataset)

Budapest.txt	Degree	0.6794	0.6709	0.4423
	Closeness	0.5126	0.5735	0.7249
	Betweenness	0.3797	0.4018	0.1675
	PageRank	0.5285	0.5381	0.2042
	Coreness	0.5272	0.6379	0.5252
	Eigenvector	0.7411	0.7357	0.7814
C_elegans.txt	Degree	0.7742	0.6321	0.7509
	Closeness	0.5081	0.5277	0.6836
	Betweenness	0.6158	0.4670	0.5792
	PageRank	0.7143	0.4915	0.6521
	Coreness	nan	0.1524	0.7707
	Eigenvector	0.7980	0.6996	0.8648
E.coli.edge	Degree	0.6608	0.6434	0.6618
	Closeness	0.5517	0.4357	0.7347
	Betweenness	0.5158	0.4524	0.3668
	PageRank	0.6495	0.5839	0.4665
	Coreness	0.5892	0.5357	0.6793
	Eigenvector	0.6963	0.6196	0.8209
Human12a.edge	Degree	0.5643	0.7351	0.8458
	Closeness	0.4251	0.3394	0.5541
	Betweenness	0.4302	0.4206	0.4746
	PageRank	0.4400	0.2945	0.5958
	Coreness	0.0531	0.1479	0.6546
	Eigenvector	0.3241	0.3127	0.5962
NewSpain_18c_travelmap.txt	Degree	0.1755	0.3952	0.6503
	Closeness	0.1996	0.3162	0.3490
	Betweenness	0.2208	0.4326	0.5239
	PageRank	0.0130	0.0909	0.3148
	Coreness	nan	0.2257	0.6016
	Eigenvector	0.4113	0.4588	0.3903
dolphins.txt	Degree	0.8281	0.6992	0.8423
	Closeness	-0.2000	0.2121	0.6008
	Betweenness	-0.2000	0.1212	0.5550
	PageRank	0.8667	0.5455	0.7205
	Coreness	nan	nan	0.7367
	Eigenvector	1.0000	0.7273	0.6337
facebook_combined.txt	Degree	0.4106	0.6368	0.7022
	Closeness	-0.1816	0.0469	0.3621
	Betweenness	-0.3730	-0.0697	0.4466
	PageRank	-0.1485	0.3191	0.3855
	Coreness	0.7394	0.7717	0.7498
	Eigenvector	0.5735	0.4154	0.4969
karate.txt	Degree	0.3333	0.7333	0.6725
	Closeness	0.3333	0.6901	0.7260
	Betweenness	1.0000	0.7333	0.5900
	PageRank	0.3333	0.7333	0.5137
	Coreness	nan	nan	0.6126
	Eigenvector	0.3333	0.8667	0.9453
mammalia-voles-bhp-trapping.edges	Degree	0.6241	0.6937	0.7595
	Closeness	0.2734	0.1850	0.4455
	Betweenness	0.4133	0.4324	0.4937
	PageRank	0.5037	0.5286	0.5117
	Coreness	0.2322	0.5164	0.7714
	Eigenvector	0.2715	0.1427	0.3911
synthetic_test_realworld.txt	Degree	0.7324	0.6763	0.5717
	Closeness	0.7395	0.6957	0.8504
	Betweenness	0.6956	0.6535	0.5930
	PageRank	0.7006	0.6396	0.4023
	Coreness	nan	nan	nan
	Eigenvector	0.7032	0.6654	0.8483
US_airports.txt	Degree	0.8232	0.8630	0.6265
	Closeness	0.6936	0.7323	0.8201
	Betweenness	0.5984	0.6063	0.3265
	PageRank	0.2637	0.3774	0.2169
	Coreness	0.5652	0.8534	0.6719
	Eigenvector	0.8661	0.9253	0.9403
cargoshipsBB.txt	Degree	0.6300	0.6706	0.7804
	Closeness	0.5174	0.6523	0.7960
	Betweenness	0.5068	0.5125	0.5839
	PageRank	0.4586	0.4458	0.5059
	Coreness	0.3844	0.6647	0.8207
	Eigenvector	0.6917	0.8085	0.9260
carrib.txt	Degree	0.8893	0.7750	0.7586
	Closeness	0.8893	0.7750	0.7586
	Betweenness	0.7391	0.7075	0.5425
	PageRank	0.3841	0.3844	0.1527
	Coreness	0.2636	-0.0084	0.8017
	Eigenvector	0.8623	0.7772	0.8901
cypedge.txt	Degree	0.8281	0.5715	0.9033
	Closeness	0.8281	0.5715	0.9033
	Betweenness	0.8667	0.5385	0.7122
	PageRank	0.6000	0.3846	0.2295
	Coreness	nan	nan	0.8938
	Eigenvector	0.8667	0.6923	0.9362
netscience.mtx	Degree	0.7406	0.6049	0.6516
	Closeness	0.4307	0.2820	0.3378
	Betweenness	0.3884	0.3735	0.3915
	PageRank	0.7132	0.5278	0.4575
	Coreness	0.6089	0.5508	0.6173
	Eigenvector	0.3902	0.0348	0.3489
open_flights.txt	Degree	0.6668	0.7339	0.5407
	Closeness	0.5035	0.5841	0.7928
	Betweenness	0.3933	0.4161	0.3722
	PageRank	0.4562	0.5450	0.2348
	Coreness	0.6669	0.8209	0.5988
	Eigenvector	0.8105	0.8238	0.8596
synthetic_test_sf_1000.txt	Degree	0.7776	0.7345	0.5777
	Closeness	0.7701	0.6880	0.8431
	Betweenness	0.7693	0.7236	0.6148
	PageRank	0.7531	0.7014	0.4205
	Coreness	nan	nan	nan
	Eigenvector	0.7451	0.6388	0.8326
synthetic_test_sf_2000.txt	Degree	0.6901	0.6444	0.5340
	Closeness	0.6744	0.6454	0.8493
	Betweenness	0.7170	0.6616	0.6331
	PageRank	0.6617	0.6127	0.3811
	Coreness	nan	nan	nan
	Eigenvector	0.5902	0.5782	0.8367
synthetic_test_sf_250.txt	Degree	0.7974	0.8237	0.6657
	Closeness	0.8227	0.8046	0.8532
	Betweenness	0.7133	0.7927	0.5725
	PageRank	0.7667	0.7959	0.4678
	Coreness	nan	nan	nan
	Eigenvector	0.8800	0.8073	0.8648
synthetic_test_sf_500.txt	Degree	0.7495	0.7339	0.6179
	Closeness	0.7864	0.7853	0.8569
	Betweenness	0.7355	0.7265	0.6020
	PageRank	0.7143	0.6998	0.4447
	Coreness	nan	nan	nan
	Eigenvector	0.7665	0.7426	0.8503
synthetic_test_sf_5000.txt	Degree	0.6102	0.5685	0.5145
	Closeness	0.6435	0.6224	0.8433
	Betweenness	0.6713	0.6271	0.6734
	PageRank	0.5889	0.5405	0.3578
	Coreness	nan	nan	nan
	Eigenvector	0.5761	0.5512	0.8175
synthetic_sf_100.txt	Degree	0.5968	0.7798	0.7882
	Closeness	0.8281	0.7669	0.8566
	Betweenness	0.3333	0.6526	0.6525
	PageRank	0.4222	0.6947	0.5996
	Coreness	nan	nan	nan
	Eigenvector	0.8222	0.8842	0.9014
synthetic_sf_1000.txt	Degree	0.7155	0.7081	0.5970
	Closeness	0.7472	0.6995	0.8396
	Betweenness	0.7624	0.7303	0.6463
	PageRank	0.6828	0.6709	0.4333
	Coreness	nan	nan	nan
	Eigenvector	0.7200	0.6571	0.8302
synthetic_sf_1500.txt	Degree	0.7150	0.6816	0.5495
	Closeness	0.7493	0.7169	0.8540
	Betweenness	0.7573	0.7156	0.6294
	PageRank	0.6918	0.6497	0.3868
	Coreness	nan	nan	nan
	Eigenvector	0.7158	0.6708	0.8452
synthetic_sf_2000.txt	Degree	0.7621	0.6303	0.5390
	Closeness	0.6891	0.6861	0.8554
	Betweenness	0.7797	0.6689	0.6399
	PageRank	0.7386	0.5970	0.3776
	Coreness	nan	nan	nan
	Eigenvector	0.6252	0.6280	0.8358
synthetic_sf_250.txt	Degree	0.7974	0.7415	0.6213
	Closeness	0.8381	0.7553	0.8457
	Betweenness	0.7667	0.6849	0.5335
	PageRank	0.7600	0.7045	0.4396
	Coreness	nan	nan	nan
	Eigenvector	0.8467	0.7747	0.8822
synthetic_sf_2500.txt	Degree	0.7073	0.6786	0.5334
	Closeness	0.7049	0.6736	0.8440
	Betweenness	0.7320	0.7068	0.6479
	PageRank	0.6822	0.6463	0.3709
	Coreness	nan	nan	nan
	Eigenvector	0.6594	0.6216	0.8279
synthetic_sf_3000.txt	Degree	0.6929	0.6553	0.5165
	Closeness	0.6737	0.5794	0.8502
	Betweenness	0.7336	0.6811	0.6437
	PageRank	0.6686	0.6251	0.3539
	Coreness	nan	nan	nan
	Eigenvector	0.6094	0.5015	0.8269
synthetic_sf_4000.txt	Degree	0.6506	0.5589	0.5142
	Closeness	0.6103	0.6472	0.8503
	Betweenness	0.6946	0.6168	0.6631
	PageRank	0.6288	0.5327	0.3565
	Coreness	nan	nan	nan
	Eig			

Table 3: Correlation (Kendall's Tau) Between SIR Spreading Capacity and Traditional Unweighted Centralities

(Light Green Highlight = Test Dataset)

Network Dataset	Centrality Metric	Top 10%	Top 20%	Top 100% (All)
Budapest.txt	Degree	0.6387	0.5944	0.5416
	Closeness	0.5468	0.5218	0.7880
	Betweenness	0.4157	0.3715	0.2573
	PageRank	0.4911	0.4818	0.2647
	Coreness	0.4444	0.5701	0.6126
	Eigenvector	0.7714	0.7306	0.8430
C_elegans.txt	Degree	0.7544	0.6618	0.7380
	Closeness	0.5428	0.5336	0.6709
	Betweenness	0.6108	0.4787	0.5666
	PageRank	0.7094	0.5289	0.6403
	Coreness	nan	0.0952	0.7574
	Eigenvector	0.8030	0.6552	0.8452
E.coli.edge	Degree	0.6211	0.5879	0.6515
	Closeness	0.5329	0.4382	0.7504
	Betweenness	0.4874	0.4257	0.3796
	PageRank	0.6041	0.5491	0.4570
	Coreness	0.5740	0.4856	0.6640
	Eigenvector	0.6812	0.6158	0.8338
Human12a.edge	Degree	0.5504	0.6251	0.7440
	Closeness	0.4277	0.3773	0.5987
	Betweenness	0.3953	0.2992	0.4035
	PageRank	0.3789	0.2540	0.5401
	Coreness	0.0122	0.3459	0.6325
	Eigenvector	0.3773	0.5162	0.7047
NewSpain_18c_travelmap.txt	Degree	0.2084	0.1903	0.2529
	Closeness	0.2082	0.4135	0.7435
	Betweenness	0.0563	0.3078	0.3303
	PageRank	0.1082	-0.0359	0.0646
	Coreness	nan	0.3137	0.4540
	Eigenvector	0.2987	0.4334	0.6145
dolphins.txt	Degree	0.6901	0.7628	0.7351
	Closeness	-0.0667	0.1818	0.7066
	Betweenness	-0.0667	0.2121	0.5116
	PageRank	0.7333	0.6364	0.6129
	Coreness	nan	nan	0.7014
	Eigenvector	0.8667	0.5758	0.7494
facebook_combined.txt	Degree	0.5002	0.7554	0.6206
	Closeness	-0.2066	-0.0472	0.4509
	Betweenness	-0.1766	0.0762	0.4219
	PageRank	-0.0316	0.4433	0.3022
	Coreness	0.7359	0.8621	0.6675
	Eigenvector	0.7489	0.6191	0.6761
karate.txt	Degree	1.0000	0.8667	0.7035
	Closeness	-0.3333	0.5521	0.7139
	Betweenness	0.3333	0.6000	0.6041
	PageRank	1.0000	0.8667	0.5653
	Coreness	nan	nan	0.6522
	Eigenvector	1.0000	1.0000	0.8729
mammalia-voles-bhp-trapping.edges	Degree	0.4512	0.3923	0.6026
	Closeness	0.2122	0.2308	0.5909
	Betweenness	0.2807	0.2269	0.4171
	PageRank	0.3698	0.3002	0.3950
	Coreness	0.1582	0.3232	0.6522
	Eigenvector	0.4768	0.4854	0.5930
synthetic_test_realworld.txt	Degree	0.7587	0.6443	0.5756
	Closeness	0.7515	0.6742	0.7822
	Betweenness	0.7354	0.6275	0.5839
	PageRank	0.7241	0.6127	0.4139
	Coreness	nan	nan	nan
	Eigenvector	0.6962	0.6347	0.7713
US_airports.txt	Degree	0.7865	0.8409	0.6527
	Closeness	0.6637	0.7144	0.7894
	Betweenness	0.5749	0.6147	0.3628
	PageRank	0.2532	0.3784	0.2384
	Coreness	0.5562	0.8273	0.6905
	Eigenvector	0.8444	0.8928	0.8755
cargoshipsBB.txt	Degree	0.6916	0.6911	0.7837
	Closeness	0.5964	0.6485	0.7835
	Betweenness	0.5574	0.5242	0.5889
	PageRank	0.4574	0.4523	0.5034
	Coreness	0.4715	0.6749	0.8212
	Eigenvector	0.8193	0.8288	0.8986
carrib.txt	Degree	0.8530	0.7391	0.7503
	Closeness	0.8530	0.7391	0.7503
	Betweenness	0.7029	0.6803	0.5447
	PageRank	0.4058	0.3605	0.1633
	Coreness	0.2636	-0.0168	0.7813
	Eigenvector	0.8406	0.7806	0.8482
cypedge.txt	Degree	0.9661	0.7273	0.9167
	Closeness	0.9661	0.7273	0.9167
	Betweenness	0.8667	0.6154	0.7201
	PageRank	0.6000	0.5128	0.2313
	Coreness	nan	nan	0.8843
	Eigenvector	0.8667	0.7692	0.9301
netscience.mtx	Degree	0.4711	0.1321	0.5143
	Closeness	0.1461	0.3066	0.3837
	Betweenness	0.1056	0.0547	0.3419
	PageRank	0.4989	0.0588	0.3346
	Coreness	0.6801	0.2467	0.4798
	Eigenvector	0.6253	0.5243	0.4802
open_flights.txt	Degree	0.6707	0.7358	0.6165
	Closeness	0.5076	0.5862	0.8064
	Betweenness	0.3993	0.4236	0.4194
	PageRank	0.4645	0.5519	0.2888
	Coreness	0.6575	0.8055	0.6651
	Eigenvector	0.7984	0.8159	0.8639
synthetic_test_sf_1000.txt	Degree	0.7722	0.6899	0.5582
	Closeness	0.7555	0.6454	0.7668
	Betweenness	0.7608	0.6755	0.5814
	PageRank	0.7455	0.6592	0.4111
	Coreness	nan	nan	nan
	Eigenvector	0.7519	0.5979	0.7538
synthetic_test_sf_2000.txt	Degree	0.6654	0.6039	0.5239
	Closeness	0.6743	0.6006	0.7694
	Betweenness	0.6892	0.6167	0.6069
	PageRank	0.6434	0.5791	0.3814
	Coreness	nan	nan	nan
	Eigenvector	0.6050	0.5382	0.7524
synthetic_test_sf_250.txt	Degree	0.7839	0.7921	0.6529
	Closeness	0.7893	0.7603	0.7728
	Betweenness	0.7000	0.7551	0.5475
	PageRank	0.7533	0.7649	0.4739
	Coreness	nan	nan	nan
	Eigenvector	0.8533	0.7731	0.7881
synthetic_test_sf_500.txt	Degree	0.7760	0.6926	0.6087
	Closeness	0.7520	0.7820	0.7794
	Betweenness	0.7339	0.6828	0.5777
	PageRank	0.7420	0.6626	0.4477
	Coreness	nan	nan	nan
	Eigenvector	0.7976	0.7838	0.7745
synthetic_test_sf_5000.txt	Degree	0.5814	0.5343	0.4712
	Closeness	0.6345	0.5775	0.7780
	Betweenness	0.6369	0.5832	0.6281
	PageRank	0.5627	0.5107	0.3334
	Coreness	nan	nan	nan
	Eigenvector	0.5627	0.5074	0.7566
synthetic_sf_100.txt	Degree	0.5013	0.6925	0.7651
	Closeness	0.6901	0.7025	0.7824
	Betweenness	0.2000	0.5684	0.6459
	PageRank	0.3778	0.6316	0.6124
	Coreness	nan	nan	nan
	Eigenvector	0.6889	0.8000	0.8169
synthetic_sf_1000.txt	Degree	0.6818	0.6649	0.5763
	Closeness	0.7347	0.6758	0.7624
	Betweenness	0.7248	0.6815	0.6052
	PageRank	0.6533	0.6317	0.4217
	Coreness	nan	nan	nan
	Eigenvector	0.7083	0.6317	0.7532
synthetic_sf_1500.txt	Degree	0.6798	0.6392	0.5391
	Closeness	0.7244	0.6684	0.7794
	Betweenness	0.7172	0.6791	0.6056
	PageRank	0.6592	0.6142	0.3879
	Coreness	nan	nan	nan
	Eigenvector	0.7015	0.6298	0.7683
synthetic_sf_2000.txt	Degree	0.7096	0.6018	0.5272
	Closeness	0.6592	0.6407	0.7804
	Betweenness	0.7288	0.6322	0.6167
	PageRank	0.6922	0.5726	0.3762
	Coreness	nan	nan	nan
	Eigenvector	0.6047	0.5880	0.7587
synthetic_sf_250.txt	Degree	0.8244	0.7382	0.6218
	Closeness	0.7780	0.7079	0.7826
	Betweenness	0.7600	0.6947	0.5264
	PageRank	0.8067	0.7045	0.4484
	Coreness	nan	nan	nan
	Eigenvector	0.7733	0.7257	0.8106
synthetic_sf_2500.txt	Degree	0.6517	0.6076	0.5063
	Closeness	0.6771	0.6417	0.7758
	Betweenness	0.6738	0.6350	0.6176
	PageRank	0.6304	0.5808	0.3587
	Coreness	nan	nan	nan
	Eigenvector	0.6413	0.5991	0.7598
synthetic_sf_3000.txt	Degree	0.6308	0.5727	0.4911
	Closeness	0.6798	0.5708	0.7871
	Betweenness	0.6851	0.6059	0.6152
	PageRank	0.6098	0.5502	0.3425
	Coreness	nan	nan	nan
	Eigenvector	0.6122	0.5003	0.7645
synthetic_sf_4000.txt	Degree	0.5903	0.5251	0.4854
	Closeness	0.5762	0.5659	0.7771
	Betweenness	0.6270	0.5758	0.6236
	PageRank	0.5724	0.5052	0.3430
	Coreness	nan	nan	nan
	Eigenvector	0.5327	0.5069	0.7610
synthetic_sf_500.txt	Degree	0.7442	0.7800	0.6462
	Closeness	0.8327	0.7328	0.7627
	Betweenness	0.7208	0.7596	0.6242
	PageRank	0.7094	0.7487	0.4828
	Coreness	nan	nan	nan
	Eigenvector	0.8041	0.7063	0.7544
synthetic_sf_850.txt	Degree	0.7318	0.6792	0.5962
	Closeness	0.7471	0.6592	0.7814
	Betweenness	0.7434	0.6736	0.6093
	PageRank	0.7003	0.6478	0.4317
	Coreness	nan	nan	nan
	Eigenvector	0.7597	0.6582	0.7804
synthetic_fragmented_1.txt	Degree	0.9129	0.4536	0.8711
	Closeness	0.6667	0.3571	0.7560
	Betweenness	0.6667	0.2143	0.5447
	PageRank	0.6667	0.2143	0.7143
	Coreness	nan	nan	0.5010
	Eigenvector	0.3333	0.6429	0.8444
synthetic_fragmented_2.txt	Degree	0.7379	0.8706	0.8530
	Closeness	0.8000	0.7957	0.8394
	Betweenness	0.6000	0.6000	0.6659
	PageRank	0.6000	0.7333	0.7082
	Coreness	nan	nan	0.6262
	Eigenvector	0.2000	0.4667	0.7961
synthetic_realworld_1.txt	Degree	0.8385	0.7775	0.6907
	Closeness	0.8463	0.7771	0.7862
	Betweenness	0.7600	0.7172	0.6048
	PageRank	0.8188	0.7451	0.5787
	Coreness	nan	nan	nan
	Eigenvector	0.8645	0.7665	0.7873
synthetic_realworld_2.txt	Degree	0.1908	0.3337	0.3807
	Closeness	0.3472	0.3914	0.6065
	Betweenness	0.2760	0.2760	0.4057
	PageRank	0.1192	0.1994	0.2193
	Coreness	nan	nan	0.0612
	Eigenvector	0.1031	0.3239	0.4413

Table 4: Correlation (Kendall's Tau) Between NLGCN Model and Weighted Centrality Measures

(Light Green Highlight = Test Dataset)

Network Dataset	Weighted Centrality Metric	Top 10%	Top 20%	Top 100% (All)
Budapest.txt	Strength (W-Deg)	0.5568	0.5446	0.2930
	Weighted Closeness	0.1478	0.1855	0.4992
	Weighted Betweenness	0.3867	0.3640	0.1728
	Weighted PageRank	0.5285	0.5381	0.2042
	Weighted Eigenvector	0.3063	0.3664	0.6770
	Strength (W-Deg)	0.7028	0.5191	0.6797
C_elegans.txt	Weighted Closeness	0.3083	0.3395	0.6107
	Weighted Betweenness	0.4089	0.3092	0.5349
	Weighted PageRank	0.7143	0.4915	0.6521
	Weighted Eigenvector	0.4236	0.4985	0.7526
	Strength (W-Deg)	0.6770	0.6277	0.6404
	Weighted Closeness	0.4357	0.3292	0.6358
	Weighted Betweenness	0.5481	0.4322	0.3909
	Weighted PageRank	0.6495	0.5839	0.4665
	Weighted Eigenvector	0.6358	0.5090	0.7504
Human12a.edge	Strength (W-Deg)	0.3910	0.2436	0.6435
	Weighted Closeness	0.2751	0.1059	0.4459
	Weighted Betweenness	0.3613	0.2128	0.3583
	Weighted PageRank	0.4400	0.2945	0.5958
	Weighted Eigenvector	0.2457	0.0881	0.3739
NewSpain_18c_travelmap.txt	Strength (W-Deg)	0.1755	0.3952	0.6503
	Weighted Closeness	0.1996	0.3162	0.3490
	Weighted Betweenness	0.2208	0.4326	0.5239
	Weighted PageRank	0.0130	0.0909	0.3148
	Weighted Eigenvector	0.4113	0.4588	0.3903
dolphins.txt	Strength (W-Deg)	0.8281	0.6992	0.8423
	Weighted Closeness	-0.2000	0.2121	0.6008
	Weighted Betweenness	-0.2000	0.1212	0.5550
	Weighted PageRank	0.8667	0.5455	0.7205
	Weighted Eigenvector	1.0000	0.7273	0.6337
facebook_combined.txt	Strength (W-Deg)	0.4106	0.6368	0.7022
	Weighted Closeness	-0.1816	0.0469	0.3621
	Weighted Betweenness	-0.3730	-0.0697	0.4466
	Weighted PageRank	-0.1485	0.3191	0.3855
	Weighted Eigenvector	0.5735	0.4154	0.4969
karate.txt	Strength (W-Deg)	0.3333	0.7333	0.6725
	Weighted Closeness	0.3333	0.6901	0.7260
	Weighted Betweenness	1.0000	0.7333	0.5900
	Weighted PageRank	0.3333	0.7333	0.5137
	Weighted Eigenvector	0.3333	0.8667	0.9453
mammalia-voles-bhp-trapping.edges	Strength (W-Deg)	0.5350	0.5652	0.6154
	Weighted Closeness	0.1942	0.2377	0.4734
	Weighted Betweenness	0.3259	0.3963	0.4564
	Weighted PageRank	0.5037	0.5286	0.5117
	Weighted Eigenvector	0.1471	0.1988	0.3204
synthetic_test_realworld.txt	Strength (W-Deg)	0.7324	0.6763	0.5717
	Weighted Closeness	0.7395	0.6957	0.8504
	Weighted Betweenness	0.6956	0.6535	0.5930
	Weighted PageRank	0.7006	0.6396	0.4023
	Weighted Eigenvector	0.7032	0.6654	0.8483
US_airports.txt	Strength (W-Deg)	-0.0220	0.2263	0.2962
	Weighted Closeness	-0.1233	0.1394	0.3782
	Weighted Betweenness	0.3258	0.2950	0.2537
	Weighted PageRank	0.2637	0.3774	0.2169
	Weighted Eigenvector	0.2163	0.2711	0.5867
cargoshipsBB.txt	Strength (W-Deg)	0.4580	0.4546	0.5518
	Weighted Closeness	0.3388	0.3182	0.3475
	Weighted Betweenness	0.4060	0.3088	0.4221
	Weighted PageRank	0.4586	0.4458	0.5059
	Weighted Eigenvector	0.3785	0.4028	0.5235
carrib.txt	Strength (W-Deg)	0.3913	0.3639	0.1336
	Weighted Closeness	0.3913	0.3690	0.1194
	Weighted Betweenness	-0.0274	0.0315	0.1524
	Weighted PageRank	0.3841	0.3844	0.1527
	Weighted Eigenvector	0.3696	0.3639	0.1239
cypedge.txt	Strength (W-Deg)	0.4667	0.3077	0.2275
	Weighted Closeness	0.4667	0.2308	0.1986
	Weighted Betweenness	0.5521	0.3844	0.3030
	Weighted PageRank	0.6000	0.3846	0.2295
	Weighted Eigenvector	0.4667	0.2821	0.2131
netscience.mtx	Strength (W-Deg)	0.7650	0.6019	0.5787
	Weighted Closeness	0.1822	0.1869	0.5194
	Weighted Betweenness	0.4636	0.4106	0.4267
	Weighted PageRank	0.7132	0.5278	0.4575
	Weighted Eigenvector	0.3065	0.3663	0.2228
open_flights.txt	Strength (W-Deg)	0.5442	0.6441	0.5220
	Weighted Closeness	0.3199	0.4179	0.6179
	Weighted Betweenness	0.3354	0.3666	0.2345
	Weighted PageRank	0.4562	0.5450	0.2348
	Weighted Eigenvector	0.5045	0.6201	0.7972
synthetic_test_sf_1000.txt	Strength (W-Deg)	0.7776	0.7345	0.5777
	Weighted Closeness	0.7701	0.6880	0.8431
	Weighted Betweenness	0.7693	0.7236	0.6148
	Weighted PageRank	0.7531	0.7014	0.4205
	Weighted Eigenvector	0.7451	0.6388	0.8326
synthetic_test_sf_2000.txt	Strength (W-Deg)	0.6901	0.6444	0.5340
	Weighted Closeness	0.6744	0.6454	0.8493
	Weighted Betweenness	0.7170	0.6616	0.6331
	Weighted PageRank	0.6617	0.6127	0.3811
	Weighted Eigenvector	0.5902	0.5782	0.8367
synthetic_test_sf_250.txt	Strength (W-Deg)	0.7974	0.8237	0.6657
	Weighted Closeness	0.8227	0.8046	0.8532
	Weighted Betweenness	0.7133	0.7927	0.5725
	Weighted PageRank	0.7667	0.7959	0.4678
	Weighted Eigenvector	0.8800	0.8073	0.8648
synthetic_test_sf_500.txt	Strength (W-Deg)	0.7495	0.7339	0.6179
	Weighted Closeness	0.7864	0.7853	0.8569
	Weighted Betweenness	0.7355	0.7265	0.6020
	Weighted PageRank	0.7143	0.6998	0.4447
	Weighted Eigenvector	0.7665	0.7426	0.8503
synthetic_test_sf_5000.txt	Strength (W-Deg)	0.6102	0.5685	0.5145
	Weighted Closeness	0.6435	0.6224	0.8433
	Weighted Betweenness	0.6713	0.6271	0.6734
	Weighted PageRank	0.5889	0.5405	0.3578
	Weighted Eigenvector	0.5761	0.5512	0.8175
synthetic_sf_100.txt	Strength (W-Deg)	0.5968	0.7798	0.7882
	Weighted Closeness	0.8281	0.7669	0.8566
	Weighted Betweenness	0.3333	0.6526	0.6525
	Weighted PageRank	0.4222	0.6947	0.5996
	Weighted Eigenvector	0.8222	0.8842	0.9014
synthetic_sf_1000.txt	Strength (W-Deg)	0.7155	0.7081	0.5970
	Weighted Closeness	0.7472	0.6995	0.8396
	Weighted Betweenness	0.7624	0.7303	0.6463
	Weighted PageRank	0.6828	0.6709	0.4333
	Weighted Eigenvector	0.7200	0.6571	0.8302
synthetic_sf_1500.txt	Strength (W-Deg)	0.7150	0.6816	0.5495
	Weighted Closeness	0.7493	0.7169	0.8540
	Weighted Betweenness	0.7573	0.7156	0.6294
	Weighted PageRank	0.6918	0.6497	0.3868
	Weighted Eigenvector	0.7158	0.6708	0.8452
synthetic_sf_2000.txt	Strength (W-Deg)	0.7621	0.6303	0.5390
	Weighted Closeness	0.6891	0.6861	0.8554
	Weighted Betweenness	0.7797	0.6689	0.6399
	Weighted PageRank	0.7386	0.5970	0.3776
	Weighted Eigenvector	0.6252	0.6280	0.8358
synthetic_sf_250.txt	Strength (W-Deg)	0.7974	0.7415	0.6213
	Weighted Closeness	0.8381	0.7553	0.8457
	Weighted Betweenness	0.7667	0.6849	0.5335
	Weighted PageRank	0.7600	0.7045	0.4396
	Weighted Eigenvector	0.8467	0.7747	0.8822
synthetic_sf_2500.txt	Strength (W-Deg)	0.7073	0.6786	0.5334
	Weighted Closeness	0.7049	0.6736	0.8440
	Weighted Betweenness	0.7320	0.7068	0.6479
	Weighted PageRank	0.6822	0.6463	0.3709
	Weighted Eigenvector	0.6594	0.6216	0.8279
synthetic_sf_3000.txt	Strength (W-Deg)	0.6929	0.6553	0.5165
	Weighted Closeness	0.6737	0.5794	0.8502
	Weighted Betweenness	0.7336	0.6811	0.6437
	Weighted PageRank	0.6686	0.6251	0.3539
	Weighted Eigenvector	0.6094	0.5015	0.8269
synthetic_sf_4000.txt	Strength (W-Deg)	0.6506	0.5589	0.5142
	Weighted Closeness	0.6103	0.6472	0.8503
	Weighted Betweenness	0.6946	0.6168	0.6631
	Weighted PageRank	0.6288	0.5327	0.3565
	Weighted Eigenvector	0.5447	0.5758	0.8329
synthetic_sf_500.txt	Strength (W-Deg)	0.7525	0.7383	0.6529
	Weighted Closeness	0.8818	0.8351	0.8478
	Weighted Betweenness	0.7518	0.7616	0.6535
	Weighted PageRank	0.7208	0.7014	0.4856
	Weighted Eigenvector	0.8514	0.8077	0.8386
synthetic_sf_850.txt	Strength (W-Deg)	0.7427	0.7181	0.5945
	Weighted Closeness	0.7830	0.6715	0.8555
	Weighted Betweenness	0.7462	0.6991	0.6234
	Weighted PageRank	0.7064	0.6784	0.4210
	Weighted Eigenvector	0.7804	0.6595	0.8592
synthetic_fragmented_1.txt	Strength (W-Deg)	0.9129	0.7835	0.8711
	Weighted Closeness	1.0000	0.6429	0.7607
	Weighted Betweenness	1.0000	0.5000	0.5494
	Weighted PageRank	1.0000	0.5000	0.7050
	Weighted Eigenvector	0.6667	0.5000	0.8630
synthetic_fragmented_2.txt	Strength (W-Deg)	0.3162	0.5804	0.8361
	Weighted Closeness	0.4000	0.7047	0.8616
	Weighted Betweenness	0.2000	0.2889	0.6470
	Weighted PageRank	0.2000	0.4222	0.6800
	Weighted Eigenvector	0.6000	0.6000	0.8243
synthetic_realworld_1.txt	Strength (W-Deg)	0.6587	0.7060	0.7104
	Weighted Closeness	0.6615	0.7856	0.8627
	Weighted Betweenness	0.6441	0.6820	0.6254
	Weighted PageRank	0.6473	0.6760	0.5852
	Weighted Eigenvector	0.6441	0.7749	0.8607
synthetic_realworld_2.txt	Strength (W-Deg)	0.2623	0.4313	0.6273
	Weighted Closeness	-0.1021	0.0570	0.1921
	Weighted Betweenness	-0.1172	0.0429	0.2003
	Weighted PageRank	0.1176	0.2479	0.4444
	Weighted Eigenvector	0.3265	0.3588	0.3985

Table 5: Correlation (Kendall's Tau) Between SIR Spreading Capacity and Weighted Centrality Measures (Light Green Highlight = Test Dataset)

Network Dataset	Weighted Centrality Metric	Top 10%	Top 20%	Top 100% (All)
Budapest.txt	Strength (W-Deg)	0.5000	0.4780	0.3789
	Weighted Closeness	0.1663	0.1864	0.5733
	Weighted Betweenness	0.4052	0.3425	0.2628
	Weighted PageRank	0.4911	0.4818	0.2647
	Weighted Eigenvector	0.2745	0.3088	0.7528
	Strength (W-Deg)	0.7078	0.5542	0.6681
C_elegans.txt	Weighted Closeness	0.3527	0.3477	0.6065
	Weighted Betweenness	0.4433	0.3442	0.5289
	Weighted PageRank	0.7094	0.5289	0.6403
	Weighted Eigenvector	0.4483	0.4775	0.7495
	Strength (W-Deg)	0.6328	0.5805	0.6322
	Weighted Closeness	0.4256	0.3428	0.6533
	Weighted Betweenness	0.5000	0.4243	0.3997
	Weighted PageRank	0.6041	0.5491	0.4570
	Weighted Eigenvector	0.6258	0.5195	0.7683
Human12a.edge	Strength (W-Deg)	0.3169	0.2475	0.6324
	Weighted Closeness	0.2140	0.1574	0.4981
	Weighted Betweenness	0.2657	0.1556	0.3190
	Weighted PageRank	0.3789	0.2540	0.5401
	Weighted Eigenvector	0.2434	0.1946	0.4527
NewSpain_18c_travelmap.txt	Strength (W-Deg)	0.2084	0.1903	0.2529
	Weighted Closeness	0.2082	0.4135	0.7435
	Weighted Betweenness	0.0563	0.3078	0.3303
	Weighted PageRank	0.1082	-0.0359	0.0646
	Weighted Eigenvector	0.2987	0.4334	0.6145
dolphins.txt	Strength (W-Deg)	0.6901	0.7628	0.7351
	Weighted Closeness	-0.0667	0.1818	0.7066
	Weighted Betweenness	-0.0667	0.2121	0.5116
	Weighted PageRank	0.7333	0.6364	0.6129
	Weighted Eigenvector	0.8667	0.5758	0.7494
facebook_combined.txt	Strength (W-Deg)	0.5002	0.7554	0.6206
	Weighted Closeness	-0.2066	-0.0472	0.4509
	Weighted Betweenness	-0.1766	0.0762	0.4219
	Weighted PageRank	-0.0316	0.4433	0.3022
	Weighted Eigenvector	0.7489	0.6191	0.6761
karate.txt	Strength (W-Deg)	1.0000	0.8667	0.7035
	Weighted Closeness	-0.3333	0.5521	0.7139
	Weighted Betweenness	0.3333	0.6000	0.6041
	Weighted PageRank	1.0000	0.8667	0.5653
	Weighted Eigenvector	1.0000	1.0000	0.8729
mammalia-voles-bhp-trapping.edges	Strength (W-Deg)	0.4257	0.3501	0.5089
	Weighted Closeness	0.1392	0.2049	0.5458
	Weighted Betweenness	0.2261	0.2274	0.3896
	Weighted PageRank	0.3698	0.3002	0.3950
	Weighted Eigenvector	-0.0091	-0.0325	0.2484
synthetic_test_realworld.txt	Strength (W-Deg)	0.7587	0.6443	0.5756
	Weighted Closeness	0.7515	0.6742	0.7822
	Weighted Betweenness	0.7354	0.6275	0.5839
	Weighted PageRank	0.7241	0.6127	0.4139
	Weighted Eigenvector	0.6962	0.6347	0.7713
US_airports.txt	Strength (W-Deg)	-0.0180	0.2382	0.3218
	Weighted Closeness	-0.1160	0.1529	0.4073
	Weighted Betweenness	0.3120	0.2886	0.2813
	Weighted PageRank	0.2532	0.3784	0.2384
	Weighted Eigenvector	0.2401	0.2903	0.5870
cargoshipsBB.txt	Strength (W-Deg)	0.4435	0.4546	0.5499
	Weighted Closeness	0.3351	0.3128	0.3481
	Weighted Betweenness	0.4060	0.3148	0.4231
	Weighted PageRank	0.4574	0.4523	0.5034
	Weighted Eigenvector	0.3616	0.3813	0.5236
carrib.txt	Strength (W-Deg)	0.4275	0.3469	0.1427
	Weighted Closeness	0.4275	0.3520	0.1278
	Weighted Betweenness	-0.0492	-0.0038	0.1547
	Weighted PageRank	0.4058	0.3605	0.1633
	Weighted Eigenvector	0.4058	0.3469	0.1294
cypedge.txt	Strength (W-Deg)	0.4667	0.4359	0.2217
	Weighted Closeness	0.4667	0.3590	0.1948
	Weighted Betweenness	0.6901	0.6639	0.3411
	Weighted PageRank	0.6000	0.5128	0.2313
	Weighted Eigenvector	0.4667	0.4103	0.2092
netscience.mtx	Strength (W-Deg)	0.6556	0.1227	0.4317
	Weighted Closeness	0.1599	0.3499	0.5570
	Weighted Betweenness	0.2531	0.1211	0.3604
	Weighted PageRank	0.4989	0.0588	0.3346
	Weighted Eigenvector	0.0497	-0.0625	0.1503
open_flights.txt	Strength (W-Deg)	0.5541	0.6485	0.5927
	Weighted Closeness	0.3307	0.4177	0.6376
	Weighted Betweenness	0.3418	0.3776	0.2825
	Weighted PageRank	0.4645	0.5519	0.2888
	Weighted Eigenvector	0.5134	0.6174	0.8057
synthetic_test_sf_1000.txt	Strength (W-Deg)	0.7722	0.6899	0.5582
	Weighted Closeness	0.7555	0.6454	0.7668
	Weighted Betweenness	0.7608	0.6755	0.5814
	Weighted PageRank	0.7455	0.6592	0.4111
	Weighted Eigenvector	0.7519	0.5979	0.7538
synthetic_test_sf_2000.txt	Strength (W-Deg)	0.6654	0.6039	0.5239
	Weighted Closeness	0.6743	0.6006	0.7694
	Weighted Betweenness	0.6892	0.6167	0.6069
	Weighted PageRank	0.6434	0.5791	0.3814
	Weighted Eigenvector	0.6050	0.5382	0.7524
synthetic_test_sf_250.txt	Strength (W-Deg)	0.7839	0.7921	0.6529
	Weighted Closeness	0.7893	0.7603	0.7728
	Weighted Betweenness	0.7000	0.7551	0.5475
	Weighted PageRank	0.7533	0.7649	0.4739
	Weighted Eigenvector	0.8533	0.7731	0.7881
synthetic_test_sf_500.txt	Strength (W-Deg)	0.7760	0.6926	0.6087
	Weighted Closeness	0.7520	0.7820	0.7794
	Weighted Betweenness	0.7339	0.6828	0.5777
	Weighted PageRank	0.7420	0.6626	0.4477
	Weighted Eigenvector	0.7976	0.7838	0.7745
synthetic_test_sf_5000.txt	Strength (W-Deg)	0.5814	0.5343	0.4712
	Weighted Closeness	0.6345	0.5775	0.7780
	Weighted Betweenness	0.6369	0.5832	0.6281
	Weighted PageRank	0.5627	0.5107	0.3334
	Weighted Eigenvector	0.5627	0.5074	0.7566
synthetic_sf_100.txt	Strength (W-Deg)	0.5013	0.6925	0.7651
	Weighted Closeness	0.6901	0.7025	0.7824
	Weighted Betweenness	0.2000	0.5684	0.6459
	Weighted PageRank	0.3778	0.6316	0.6124
	Weighted Eigenvector	0.6889	0.8000	0.8169
synthetic_sf_1000.txt	Strength (W-Deg)	0.6818	0.6649	0.5763
	Weighted Closeness	0.7347	0.6758	0.7624
	Weighted Betweenness	0.7248	0.6815	0.6052
	Weighted PageRank	0.6533	0.6317	0.4217
	Weighted Eigenvector	0.7083	0.6317	0.7532
synthetic_sf_1500.txt	Strength (W-Deg)	0.6798	0.6392	0.5391
	Weighted Closeness	0.7244	0.6684	0.7794
	Weighted Betweenness	0.7172	0.6791	0.6056
	Weighted PageRank	0.6592	0.6142	0.3879
	Weighted Eigenvector	0.7015	0.6298	0.7683
synthetic_sf_2000.txt	Strength (W-Deg)	0.7096	0.6018	0.5272
	Weighted Closeness	0.6592	0.6407	0.7804
	Weighted Betweenness	0.7288	0.6322	0.6167
	Weighted PageRank	0.6922	0.5726	0.3762
	Weighted Eigenvector	0.6047	0.5880	0.7587
synthetic_sf_250.txt	Strength (W-Deg)	0.8244	0.7382	0.6218
	Weighted Closeness	0.7780	0.7079	0.7826
	Weighted Betweenness	0.7600	0.6947	0.5264
	Weighted PageRank	0.8067	0.7045	0.4484
	Weighted Eigenvector	0.7733	0.7257	0.8106
synthetic_sf_2500.txt	Strength (W-Deg)	0.6517	0.6076	0.5063
	Weighted Closeness	0.6771	0.6417	0.7758
	Weighted Betweenness	0.6738	0.6350	0.6176
	Weighted PageRank	0.6304	0.5808	0.3587
	Weighted Eigenvector	0.6413	0.5991	0.7598
synthetic_sf_3000.txt	Strength (W-Deg)	0.6308	0.5727	0.4911
	Weighted Closeness	0.6798	0.5708	0.7871
	Weighted Betweenness	0.6851	0.6059	0.6152
	Weighted PageRank	0.6098	0.5502	0.3425
	Weighted Eigenvector	0.6122	0.5003	0.7645
synthetic_sf_4000.txt	Strength (W-Deg)	0.5903	0.5251	0.4854
	Weighted Closeness	0.5762	0.5659	0.7771
	Weighted Betweenness	0.6270	0.5758	0.6236
	Weighted PageRank	0.5724	0.5052	0.3430
	Weighted Eigenvector	0.5327	0.5069	0.7610
synthetic_sf_500.txt	Strength (W-Deg)	0.7442	0.7800	0.6462
	Weighted Closeness	0.8327	0.7328	0.7627
	Weighted Betweenness	0.7208	0.7596	0.6242
	Weighted PageRank	0.7094	0.7487	0.4828
	Weighted Eigenvector	0.8041	0.7063	0.7544
synthetic_sf_850.txt	Strength (W-Deg)	0.7318	0.6792	0.5962
	Weighted Closeness	0.7471	0.6562	0.7814
	Weighted Betweenness	0.7434	0.6736	0.6093
	Weighted PageRank	0.7003	0.6478	0.4317
	Weighted Eigenvector	0.7597	0.6582	0.7804
synthetic_fragmented_1.txt	Strength (W-Deg)	0.9129	0.4536	0.8711
	Weighted Closeness	0.6667	0.3571	0.7560
	Weighted Betweenness	0.6667	0.2143	0.5447
	Weighted PageRank	0.6667	0.2143	0.7143
	Weighted Eigenvector	0.3333	0.6429	0.8444
synthetic_fragmented_2.txt	Strength (W-Deg)	0.7379	0.8706	0.8530
	Weighted Closeness	0.8000	0.7957	0.8394
	Weighted Betweenness	0.6000	0.6000	0.6659
	Weighted PageRank	0.6000	0.7333	0.7082
	Weighted Eigenvector	0.2000	0.4667	0.7961
synthetic_realworld_1.txt	Strength (W-Deg)	0.8385	0.7775	0.6907
	Weighted Closeness	0.8463	0.7771	0.7862
	Weighted Betweenness	0.7600	0.7172	0.6048
	Weighted PageRank	0.8188	0.7451	0.5787
	Weighted Eigenvector	0.8645	0.7665	0.7873
synthetic_realworld_2.txt	Strength (W-Deg)	0.1908	0.3337	0.3807
	Weighted Closeness	0.3472	0.3914	0.6065
	Weighted Betweenness	0.2760	0.2760	0.4057
	Weighted PageRank	0.2192	0.1994	0.2193
	Weighted Eigenvector	0.1031	0.3239	0.4413

Table 6: Correlation (Kendall's Tau) Between Traditional Unweighted and Weighted Centrality Measures (Light Green Highlight = Test Dataset)

Network Dataset	Centrality Comparison	Top 10%	Top 20%	Top 100% (All)
Budapest.txt	Degree vs Strength	0.7829	0.7853	0.8233
	Closeness (Unw vs W)	0.4428	0.3241	0.6218
	Betweenness (Unw vs W)	0.6825	0.5997	0.8008
	PageRank (Unw vs W)	1.0000	1.0000	1.0000
C_elegans.txt	Eigenvector (Unw vs W)	0.1865	0.2116	0.8117
	Degree vs Strength	0.6733	0.7659	0.8093
	Closeness (Unw vs W)	0.1687	0.2613	0.4550
	Betweenness (Unw vs W)	0.3498	0.5196	0.6094
E.coli.edge	PageRank (Unw vs W)	1.0000	1.0000	1.0000
	Eigenvector (Unw vs W)	0.5567	0.5535	0.7450
	Degree vs Strength	0.8958	0.8909	0.9194
	Closeness (Unw vs W)	0.6832	0.6921	0.7846
Human12a.edge	Betweenness (Unw vs W)	0.8055	0.7910	0.7721
	PageRank (Unw vs W)	1.0000	1.0000	1.0000
	Eigenvector (Unw vs W)	0.8121	0.7852	0.8835
	Degree vs Strength	0.3654	0.2847	0.6516
NewSpain_18c_travelmap.txt	Closeness (Unw vs W)	0.2776	0.2504	0.5194
	Betweenness (Unw vs W)	0.3498	0.2678	0.4387
	PageRank (Unw vs W)	1.0000	1.0000	1.0000
	Eigenvector (Unw vs W)	0.6473	0.5305	0.6681
dolphins.txt	Degree vs Strength	1.0000	1.0000	1.0000
	Closeness (Unw vs W)	1.0000	1.0000	1.0000
	Betweenness (Unw vs W)	1.0000	1.0000	1.0000
	PageRank (Unw vs W)	1.0000	1.0000	1.0000
facebook_combined.txt	Eigenvector (Unw vs W)	1.0000	1.0000	1.0000
	Degree vs Strength	1.0000	1.0000	1.0000
	Closeness (Unw vs W)	1.0000	1.0000	1.0000
	Betweenness (Unw vs W)	1.0000	1.0000	1.0000
karate.txt	PageRank (Unw vs W)	1.0000	1.0000	1.0000
	Eigenvector (Unw vs W)	1.0000	1.0000	1.0000
	Degree vs Strength	1.0000	1.0000	1.0000
	Closeness (Unw vs W)	1.0000	1.0000	1.0000
mammalia-voles-bhp-trapping.edges	Betweenness (Unw vs W)	1.0000	1.0000	1.0000
	PageRank (Unw vs W)	1.0000	1.0000	1.0000
	Eigenvector (Unw vs W)	1.0000	1.0000	1.0000
	Degree vs Strength	0.6426	0.6710	0.7458
synthetic_test_realworld.txt	Closeness (Unw vs W)	0.6013	0.4959	0.5768
	Betweenness (Unw vs W)	0.6569	0.6791	0.7298
	PageRank (Unw vs W)	1.0000	1.0000	1.0000
	Eigenvector (Unw vs W)	-0.4052	-0.4021	-0.0460
US_airports.txt	Degree vs Strength	1.0000	1.0000	1.0000
	Closeness (Unw vs W)	1.0000	1.0000	1.0000
	Betweenness (Unw vs W)	1.0000	1.0000	1.0000
	PageRank (Unw vs W)	1.0000	1.0000	1.0000
cargoshipsBB.txt	Eigenvector (Unw vs W)	0.2098	0.2473	0.6025
	Degree vs Strength	0.5532	0.5394	0.5493
	Closeness (Unw vs W)	0.3154	0.2892	0.3776
	Betweenness (Unw vs W)	0.5499	0.4790	0.5450
carrib.txt	PageRank (Unw vs W)	1.0000	1.0000	1.0000
	Eigenvector (Unw vs W)	0.4026	0.4259	0.5407
	Degree vs Strength	0.3376	0.3422	0.0946
	Closeness (Unw vs W)	0.3376	0.3507	0.0852
cypedge.txt	Betweenness (Unw vs W)	0.1040	0.1198	0.2128
	PageRank (Unw vs W)	1.0000	1.0000	1.0000
	Eigenvector (Unw vs W)	0.3478	0.3350	0.1451
	Degree vs Strength	0.4140	0.3896	0.2380
netscience.mtx	Closeness (Unw vs W)	0.4140	0.3117	0.2117
	Betweenness (Unw vs W)	0.5521	0.5591	0.3344
	PageRank (Unw vs W)	1.0000	1.0000	1.0000
	Eigenvector (Unw vs W)	0.3333	0.2308	0.2246
open_flights.txt	Degree vs Strength	0.6351	0.6101	0.8069
	Closeness (Unw vs W)	0.3130	0.5531	0.5795
	Betweenness (Unw vs W)	0.5955	0.6330	0.6426
	PageRank (Unw vs W)	1.0000	1.0000	1.0000
synthetic_test_sf_1000.txt	Eigenvector (Unw vs W)	-0.1791	-0.4018	-0.2816
	Degree vs Strength	0.8084	0.8667	0.9167
	Closeness (Unw vs W)	0.5300	0.5183	0.6766
	Betweenness (Unw vs W)	0.7525	0.7735	0.7588
synthetic_test_sf_2000.txt	PageRank (Unw vs W)	1.0000	1.0000	1.0000
	Eigenvector (Unw vs W)	0.4950	0.5770	0.8466
	Degree vs Strength	1.0000	1.0000	1.0000
	Closeness (Unw vs W)	1.0000	1.0000	1.0000
synthetic_test_sf_500.txt	Betweenness (Unw vs W)	1.0000	1.0000	1.0000
	PageRank (Unw vs W)	1.0000	1.0000	1.0000
	Eigenvector (Unw vs W)	1.0000	1.0000	1.0000
	Degree vs Strength	1.0000	1.0000	1.0000
synthetic_test_sf_5000.txt	Closeness (Unw vs W)	1.0000	1.0000	1.0000
	Betweenness (Unw vs W)	1.0000	1.0000	1.0000
	PageRank (Unw vs W)	1.0000	1.0000	1.0000
	Eigenvector (Unw vs W)	1.0000	1.0000	1.0000
synthetic_sf_100.txt	Degree vs Strength	1.0000	1.0000	1.0000
	Closeness (Unw vs W)	1.0000	1.0000	1.0000
	Betweenness (Unw vs W)	1.0000	1.0000	1.0000
	PageRank (Unw vs W)	1.0000	1.0000	1.0000
synthetic_sf_1000.txt	Eigenvector (Unw vs W)	1.0000	1.0000	1.0000
	Degree vs Strength	1.0000	1.0000	1.0000
	Closeness (Unw vs W)	1.0000	1.0000	1.0000
	Betweenness (Unw vs W)	1.0000	1.0000	1.0000
synthetic_sf_1500.txt	PageRank (Unw vs W)	1.0000	1.0000	1.0000
	Eigenvector (Unw vs W)	1.0000	1.0000	1.0000
	Degree vs Strength	1.0000	1.0000	1.0000
	Closeness (Unw vs W)	1.0000	1.0000	1.0000
synthetic_sf_2000.txt	Betweenness (Unw vs W)	1.0000	1.0000	1.0000
	PageRank (Unw vs W)	1.0000	1.0000	1.0000
	Eigenvector (Unw vs W)	1.0000	1.0000	1.0000
	Degree vs Strength	1.0000	1.0000	1.0000
synthetic_sf_250.txt	Closeness (Unw vs W)	1.0000	1.0000	1.0000
	Betweenness (Unw vs W)	1.0000	1.0000	1.0000
	PageRank (Unw vs W)	1.0000	1.0000	1.0000
	Eigenvector (Unw vs W)	1.0000	1.0000	1.0000
synthetic_sf_2500.txt	Degree vs Strength	1.0000	1.0000	1.0000
	Closeness (Unw vs W)	1.0000	1.0000	1.0000
	Betweenness (Unw vs W)	1.0000	1.0000	1.0000
	PageRank (Unw vs W)	1.0000	1.0000	1.0000
synthetic_sf_3000.txt	Eigenvector (Unw vs W)	1.0000	1.0000	1.0000
	Degree vs Strength	1.0000	1.0000	1.0000
	Closeness (Unw vs W)	1.0000	1.0000	1.0000
	Betweenness (Unw vs W)	1.0000	1.0000	1.0000
synthetic_sf_4000.txt	PageRank (Unw vs W)	1.0000	1.0000	1.0000
	Eigenvector (Unw vs W)	1.0000	1.0000	1.0000
	Degree vs Strength	1.0000	1.0000	1.0000
	Closeness (Unw vs W)	1.0000	1.0000	1.0000
synthetic_sf_500.txt	Betweenness (Unw vs W)	1.0000	1.0000	1.0000
	PageRank (Unw vs W)	1.0000	1.0000	1.0000
	Eigenvector (Unw vs W)	1.0000	1.0000	1.0000
	Degree vs Strength	1.0000	1.0000	1.0000
synthetic_sf_850.txt	Closeness (Unw vs W)	1.0000	1.0000	1.0000
	Betweenness (Unw vs W)	1.0000	1.0000	1.0000
	PageRank (Unw vs W)	1.0000	1.0000	1.0000
	Eigenvector (Unw vs W)	1.0000	1.0000	1.0000
synthetic_fragmented_1.txt	Degree vs Strength	1.0000	1.0000	1.0000
	Closeness (Unw vs W)	1.0000	1.0000	1.0000
	Betweenness (Unw vs W)	1.0000	1.0000	1.0000
	PageRank (Unw vs W)	1.0000	1.0000	1.0000
synthetic_fragmented_2.txt	Eigenvector (Unw vs W)	1.0000	1.0000	1.0000
	Degree vs Strength	1.0000	1.0000	1.0000
	Closeness (Unw vs W)	1.0000	1.0000	1.0000
	Betweenness (Unw vs W)	1.0000	1.0000	1.0000
synthetic_realworld_1.txt	PageRank (Unw vs W)	1.0000	1.0000	1.0000
	Eigenvector (Unw vs W)	1.0000	1.0000	1.0000
	Degree vs Strength	1.0000	1.0000	1.0000
	Closeness (Unw vs W)	1.0000	1.0000	1.0000
synthetic_realworld_2.txt	Betweenness (Unw vs W)	1.0000	1.0000	1.0000
	PageRank (Unw vs W)	1.0000	1.0000	1.0000
	Eigenvector (Unw vs W)	1.0000	1.0000	1.0000
	Degree vs Strength	1.0000	1.0000	1.0000